BRNO UNIVERSITY OF TECHNOLOGY
Faculty of Information Technology

Network Applications and Network Administration (ISA)

Whois supercalifragilisticexpialidocious whisperer

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Filip Kuchař

1. Introduction

This project provides an implementation of Whois supercalifragilisticexpialidocious whisperer (isa-tazatel) tool for the Network Applications and Network Administration (ISA) subject.

isa-tazatel is a whois and a DNS lookup tool. This document describes the implementation, future overview and how to use it.

2. Vocabulary

2.1 DNS

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).[5]

2.2 Whois

WHOIS (pronounced as the phrase "who is") is a query and response protocol that is widely used for querying databases that store the registered users or assignees of an Internet resource, such as a domain name, an IP address block or an autonomous system, but is also used for a wider range of other information. The protocol stores and delivers database content in a human-readable format.[4]

3. Design and Implementation

C++17 was chosen as the go-to language due to its OOP abilities.

3.1 Parsing the cmd options

check_argv handles the parsing, using getopt_long as the foundation.

3.2 DNS

The DNS part relies heavily on the linux resolver library (resolv.h) for creating, sending, receiving and parsing the DNS query. Thread-safe variants of function calls are used due to the multithreaded nature of this application - a worker for each query.

Support for setting arbitrary DNS server is implemented also by means of resolv.h, although with couple of catches. Some systems do not support res_setserver routine for setting a dns server but expose inner extension structures allowing setting server directly.

Function set_dnsserver wrapps that ability and is called once per dns_query task as it takes res_state.

This app also provides a form of recursion to maximize the use of fetched DNS information.
dns_query takes a Time to Live (TTL) parameter used to limit the number of recurring queries.
TTL is not user-definable.
The figure below shows an example of the resulting structure of obtained DNS information.

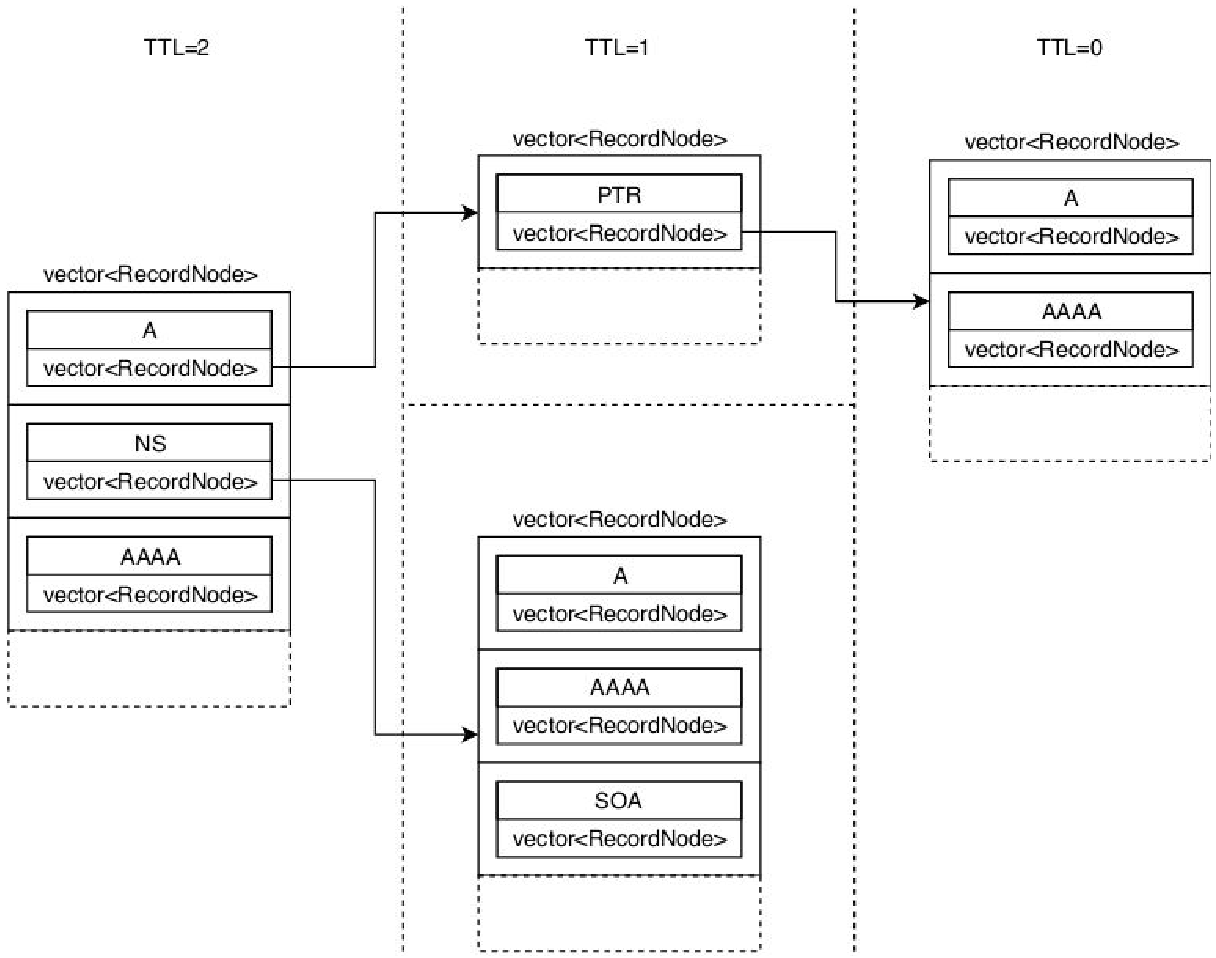

figure 1

3.2 Whois

Whois module uses BSD sockets for communication and corresponding system calls are used e.g. create, send and recv. Timeout for connect & recv is set to 5s. It has dependency on dns module because if the user specifies whois server as a hostname, we need to resolve it. Same applies to a query.

For whois queries, IP address is used followed by “\r\n” to comply with the whois query format[1]. Then check is done if the response contains a redirection string in case the issued server just points to a different server that might have the answer.

Due to the various formats that a whois response might have, if it does not contain a redirection information, the response is printed on the standard output. Because responses can be very long, no recursion is implemented for whois querying.

4. Instructions for Use

4.1 Requirements

You will need the Linux Resolver library, GNU Make and GCC 7
Tested on:
Ubuntu 18.04.3 LTS
Fedora 30

4.2 Compilation

make compiling the project with O2 optimization enabled.
make debug enabling debug printing info, file:line output during memory leak tracking with Valgrind
Eventually producing *isa-tazatel* binary in the project's root directory.
make clean removing obj files

4.3 Usage

Synopsis

isa-tazatel [-h] [-d server] -q host -w whost

Options

-d <u>server</u>

Server can be either IPv4 or IPv6.
If -d is not specified, the system default DNS server is used.

-h

Print (on the standard output) a description of the command-line options understood by isa-tazatel

-d <u>server</u>

Use server as a DNS for querying instead of the system default
A value for this option must be provided; possible formats are: IPv4, IPv6
If server not specified, system default DNS is used.

-q <u>host</u>

Use host as a DNS and Whois query.
Regarding Whois query, hostname gets automatically resolved nad IP is used as a query.
A value for this option must be provided; possible formats are: hostname, IPv4, IPV6

-w whost

whost is the name or IP address of the whois server to query
A value for this option must be provided; possible formats are: hostname, IPv4, IPV6

5. References and Literature

[1] Whois query format
https://tools.ietf.org/html/rfc3912#section-3
[2] Inventory and Analysis of WHOIS Registration Objects
https://tools.ietf.org/html/rfc7485
[3] DNS Messages
http://www.zytrax.com/books/dns/ch15
[4] What is DNS?
https://www.cloudflare.com/learning/dns/what-is-dns
[5] Whois RFC 3912
https://tools.ietf.org/html/rfc3912